COURSE 2 · SECTION 7 · WORKED SOLUTION

Two dates, one bridge

Section 7 (videos S07.1 and S07.2) · Data model + DAX for engineering data

DESIGN QUESTION

How many changes were requested and how many were closed in each year, and how many changes touch the P300 seat ring: counted by primary part, and counted by every part a change affects?

Steps

1. Open Section08_1_Start.pbix (or Engxtech_S07_2.pbix) > File > Save as > Engxtech_S07_Exercise.
2. Model view: hover each line into Changes; the CloseDate and PrimaryPart lines are dashed (inactive).
3. Matrix: rows Dates[Year]; values Changes Requested, Changes Closed. Matrix: rows Parts[PartID]; values Changes Affecting Part, Changes by Primary Part.
4. Right-click _Measures > New measure: Primary Part Changes (TREATAS) (below). Add it to matrix 2.
5. Cards: Closed %; a table visual with ChangeAffectedItems[ChangeID] set to Count shows the bridge rows.

DAX calculated table (from the video), then Mark as date table

```
Dates =
ADDCOLUMNS (
    CALENDAR ( DATE ( 2024, 1, 1 ), DATE ( 2026, 12, 31 ) ),
    "Year", YEAR ( [Date] ),
    "MonthNo", MONTH ( [Date] ),
    "Month", FORMAT ( [Date], "MMM" ),
    "YearMonth", FORMAT ( [Date], "YYYY-MM" ),
    "Quarter", "Q" & QUARTER ( [Date] )
)
```

DAX measures (_Measures)

```
Change Count      = COUNTROWS ( Changes )
Changes Requested = [Change Count]
Changes Closed =
CALCULATE ( [Change Count],
    USERELATIONSHIP ( Changes[CloseDate], Dates[Date] ),
    NOT ISBLANK ( Changes[CloseDate] ) )
Closed % = DIVIDE ( [Changes Closed], [Change Count] )

Changes Affecting Part  = DISTINCTCOUNT ( ChangeAffectedItems[ChangeID] )
Changes by Primary Part =
CALCULATE ( [Change Count], USERELATIONSHIP ( Changes[PrimaryPart], Parts[PartID] ) )

-- new in this exercise
Primary Part Changes (TREATAS) =
CALCULATE ( [Change Count], TREATAS ( VALUES ( Parts[PartID] ), Changes[PrimaryPart] ) )
```

USERELATIONSHIP switches the Dates filter from RequestDate to CloseDate for this one measure, so Dates[Year] = 2025 counts changes closed in 2025. Changes that are not closed have no CloseDate; NOT ISBLANK keeps them out even when no date filter is applied (a card shows 154, not 178). DIVIDE returns blank instead of an error when the denominator is 0.

The two part counts answer different questions. Changes by Primary Part counts each change once, under its main part (P300 26). Changes Affecting Part counts every change that touches the part through the bridge (P300 45), so the part rows add up to more than 178. Say which one you show. TREATAS and USERELATIONSHIP give the same primary-part count here; USERELATIONSHIP needs the inactive line, TREATAS needs no line at all.

Expected result

Check	Result
Change Count, all	**178**
Requested in 2025 / 2026	**81 / 97**
Closed in 2025 / 2026 (USERELATIONSHIP)	**66 / 88**
Closed %	**86.5 %**
P300: Changes Affecting Part (bridge)	**45**
P300: Changes by Primary Part	**26**
ChangeAffectedItems rows	**304**